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Explicit Sparse Self-Attentive Network for CTR Prediction

Yu Luo, Wanwan Peng, Youping Fan, Hong Pang, Xiang Xu, Xiaohua Wu*

School of Information and Software Engineering, University of Electronic Science and Technology of China, Chengdu, 610054, China

Abstract

The click-through rate (CTR) prediction is an essential task in the recommendation system. Available raw category features and feature combinations are important for CTR prediction, but their sparsity and high-dimension lead to huge computation. In recent years, methods based on deep learning, factorization machines and attention mechanism are widely used to model feature interactions among features for effective sparse predictive analysis. However, these previous algorithms are affected by irrelevant information. In this paper, we propose a novel model ESAtInt to model high-dimensional and sparse features, while being able to explicitly select meaningful higher-order feature interactions and eliminate the impact of irrelevant information. Specifically, we further conduct automatic feature interaction learning through a concentrated attention mechanism with explicit selection. Our model can explicitly select the most relevant parts to emphasize the attention to global features. Experimental results on the real-world dataset show that our proposed method is superior to state-of-the-art approaches for CTR prediction.

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Keywords: Concentrated attention; CTR prediction; Feature interactions.

1. Introduction

How to effectively model feature interactions and gain meaningful high-order combinational features is still an ongoing research topic in CTR prediction.

In recent years, deep learning has gradually been applied to CTR prediction task. FM⁷ improved the linear model by using factorization technology. Subsequently, different variants of factorization machines were proposed. FFM⁶ introduced the concept of field to carry out fine-grained feature interactions. AFM¹⁰ introduced the attention mechanism. All the above models focus on modeling of low-order feature interactions. Wide&Deep¹ combines linear models with neural networks. DeepFM³ connects neural networks and factorization machine in parallel. The

* Corresponding E-mail address: wuxh@uestc.edu.cn

methods mentioned above have been widely applied in CTR tasks, however, they usually suffer from some limitations. It's a challenging task to model meaningful and interpretable high-order feature interactions. Recently, some studies⁸ apply the attention mechanism to automatic feature interaction, which can model feature interactions to a certain extent. Since the attention mechanism⁹ distributes credit to all components of the feature interactions. The attention assigns a high reputation to many unrelated feature interactions.

We propose a method based on the multi-head explicit selection self-attention mechanism¹¹, which can effectively model feature combinations of different orders and reduce the interference of irrelevant information. Inspired by the success of Explicit Sparse Transformer¹¹ in the field of natural language processing, we utilize the concentrated attention mechanism to learn the weights of feature interactions dynamically. Specifically, we design a novel interaction layer to concentrate attention on the most relevant k feature interactions. In each interaction layer, each feature can interact with the most relevant k features through explicit selection, and can automatically identify related features through a multi-head attention mechanism to form meaningful high-order features.

The contributions of this study are as follows: (i) We propose a novel model called Explicit Sparse Attention Interaction (ESAtInt) based on concentrated self-attentive neural networks, which can explicitly select the most relevant high-order feature interactions, thus effectively processing large-scale structured sparse data. (ii) We conducted extensive experiments on the real-world datasets. The experiment results demonstrate the superiority of our proposed ESAtInt compared with other state-of-the-art models in the CTR prediction task.

2. Method

Our method is dedicated to handle sparse high-dimensional features and capture the meaningful combination data through an explicit selection of self-attention. Our model structure is shown in Fig. 1.

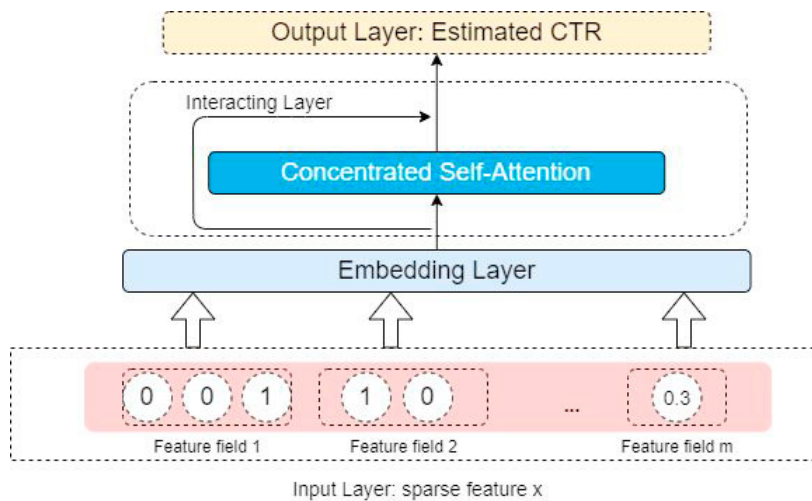


Fig. 1. Overview of model ESAtInt.

2.1. Input Layer

Our model takes multi-field categorical features as input, i.e., $x = [x_{field1}; x_{field2}; \dots; x_{fieldm}]$, where x_{fieldi} is the field-aware one-hot encoding of the i -th field, and m indicates the total number of feature fields.

2.2. Embedding Layer

The embedding layer maps each original feature into a dense vector representation. The output of embedding layer is a concatenated vector $e = [e_1, e_2, \dots, e_m]$, where m indicates the number of fields, and $e_i \in R^D$ denotes the embedding of the i -th field.

2.3. Interacting Layer

Next, we start to model low-order and high-order feature interactions in low-dimensional vector spaces. The critical problem is to select features that can be combined into meaningful feature interactions. The self-attentive network has demonstrated most advanced performances in CTR prediction tasks. Self-attention¹² is able to simulate high-order feature interactions, but it may be affected by extracting irrelevant information. In this paper, we solve this problem with a novel interacting layer as presented in Fig. 2, which use the explicit sparse multi-head self-attention mechanism¹⁵.

First, the attention score matrix $\mathbf{P}$ is generated under a specific attention head h , which represents the correlation between each feature and other features,

$$P = \frac{QK^T}{\sqrt{d}} \quad (1)$$

Where $Q = W_Q^{(h)}e$ and $K = W_K^{(h)}e$ are the linear transformation of the feature vectors, $W_Q^{(h)}, W_K^{(h)} \in R^{d' \times d}$ are transformation matrices.

The difference between our proposed model and the previous model rests with the realization of self-attention. With $top-k$ selection, attention will degenerate into sparse attention. In this way, k feature interactions that contribute the most to attention are preserved, while other irrelevant information is deleted. We hypothesis that the value of score P_{ij} with larger values demonstrate that feature i and feature j have a higher relevance. Therefore, in order to select $top-k$ contributed feature interactions, we utilize the following masking operation $M(\cdot, \cdot)$ to model feature interactions:

$$M(P, k)_{ij} = \begin{cases} P_{ij} & P_{ij} \geq t_i \\ -\infty & P_{ij} \leq t_i \end{cases} \quad (2)$$

where t_i is the k -th largest value of row i of matrix $\mathbf{P}$. Specifically, we select and record the position (i, j) of the k largest elements in each row of $\mathbf{P}$, where k is a hyperparameter. If the value of the j -th element in row i is greater than the k -th maximum value t_i , record the position (i, j) , $[t_1, t_2, \dots, t_d]$ is the vector that composed of thresholds.

In our model, $top-k$ selection is used to select the high-order combinational features with high attention score in an explicit way, which is different from the general sparse attention. The latter does not consider its importance and randomly gives up some feature interactions. Next, the masking matrix is normalized as $A = softmax(M(P, k))$, and under the guidance of $\mathbf{A}$, the feature representation in subspace h can be updated by combining all relevant features through mask operation, i.e., $C^{(h)} = AV$ and $V = W_V^{(h)}e$.

The combinatorial features collected by our model in all head subspaces as follows:

$$\tilde{C} = C^{(1)} \oplus C^{(2)} \oplus \dots \oplus C^{(H)} \quad (3)$$

where $\oplus$ is the concatenation operator, and $\mathbf{H}$ is the number of total heads.

Then the standard residual connections⁴ are added to keep the raw individual features and low-order feature interactions.

$$C^{Res} = \text{ReLU}(\tilde{C} + W_{Res}V) \quad (4)$$

Two general metrics: (1) AUC is the criterion for judging the pros and cons of the binary classification prediction model. (2) Since minimizing the Logloss defined by equation 6 is the goal of all models, it can be used as a straight forward metric. It is worth noting that for the CTR prediction task, improving AUC or decreasing Logloss even 0.001 is considered important, which has also been pointed out in existing work.

We compared with 5 representative models designed for sparse data prediction: FM⁹, NFM⁵, DeepFM³, AFM¹³ and AutoInt¹¹.

All methods are implemented in TensorFlow with embedding dimension of 8 and batch size of 128. All models based on deep neural network are optimized by using Adam. To prevent overfitting, the dropout ratio is set to 0.5. It is worth mentioning that we stack three interacting layers in following experiments and use two attention heads in each layer.

3.1. Performance Comparison (RQ1)

In this subsection, we compare the effectiveness of different algorithms on the test set. The best performance each method after multiple tests is summarized in Table 1. The results in Table 1 shows that our method achieves the best performance among all baseline methods. Our model is superior to AutoInt according to AUC evaluation. Note that in addition to the self-attention network, the structure of our model is basically the same as AutoInt, which indicates that it is crucial to utilize explicit selection attention mechanism to learn meaningful feature interactions.

Table 1. Performance comparison of different algorithms.

Method	AUC	Logloss
FM	0.6255	0.1538
NFM	0.6286	0.1536
DeepFM	0.6295	0.1531
AFM	0.6290	0.1538
AutoInt	0.6317	0.1528
ESAtInt (Ours)	0.6325	0.1528

3.2. Hyper-parameter Investigation (RQ2)

As shown in Fig. 3, we compare the influence of the value of k on model performance, and set $k \in \{4, 8, 16, 32\}$. We can see that with the increase of k , the model performance increases gradually. When k reaches 16, the performance is optimal, and then decreases instead. Specially, “inf” denotes the special case where all positions may be attended, same as the origin attention.

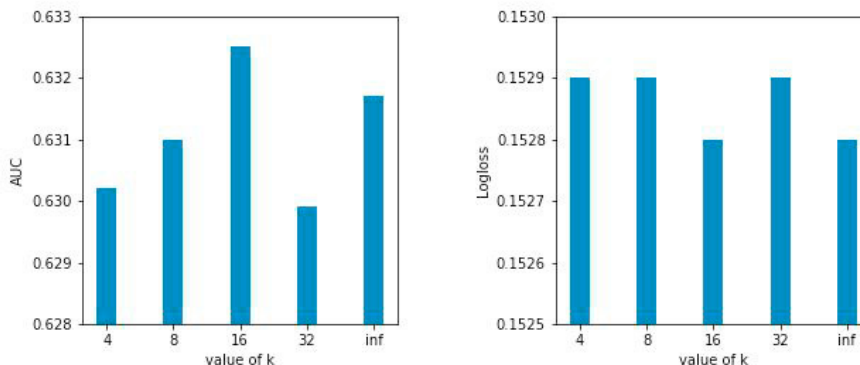


Fig. 3. Analyze the value of k on Safe Driver Prediction datasets.

3.3. Impact of the Concentrated Attention Network (RQ3)

Next, we center on analyzing the impact of the concentrated attention on our model. To investigate whether concentrated attention network further enhances the performance, we combine other two types attention interacting layers with two layers of feed-forward neural network through joint training respectively. The other attention mechanisms are: base attention and sparse attention². Specially, the value of sparse attention matrix is set to zero except for the attention matrix positions with a relative distance of not more than k and a relative distance of $k, 2k, 3k, \dots, nk$.

The evaluation results of joint-training models are presented in Table 2, indicating that concentrated attention mechanism does improve the predictive capability of the model.

Table 2. Results of the three types attention mechanism networks.

Method	AUC	Logloss
Base	0.6317	0.1529
Ours	0.6325	0.1528
Sparse	0.6302	0.1529

4. Conclusions

In this paper, we discuss the influence of high-order feature interactions on CTR prediction. In order to overcome the shortcomings of existing model, we propose a novel model ESAtInt based on concentrated attention. Our ESAtInt explicitly selects meaningful high-order feature interactions, eliminating the influence of irrelevant information. On real-world datasets, we get better off-line AUC and Logloss than competition methods. Experimental results demonstrate the validity of our model. In future work, we will explore the application of the concentrated attention mechanism for sequential recommendation. We will consider capturing effective sequence information to improve the performance of temporal predictive analytics model. Another direction that interest us is to develop ESAtInt variants based on knowledge graph to improve the explainable of the model.

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References

1. Cheng H, Koc L, Harmsen J, et al. Wide & Deep Learning for Recommender Systems. conference on recommender systems, 2016: 7-10.
2. Child R, Gray S, Radford A, et al. Generating Long Sequences with Sparse Transformers.. arXiv: Learning, 2019.
3. Guo H, Tang R, Ye Y, et al. DeepFM: A Factorization-Machine based Neural Network for CTR Prediction. arXiv: Information Retrieval, 2017.
4. He K, Zhang X, Ren S, et al. Deep Residual Learning for Image Recognition. computer vision and pattern recognition, 2016: 770-778.
5. He X, Chua T. Neural Factorization Machines for Sparse Predictive Analytics. international acm sigir conference on research and development in information retrieval, 2017: 355-364.
6. Juan Y, Zhuang Y, Chin W, et al. Field-aware Factorization Machines for CTR Prediction. conference on recommender systems, 2016: 43-50.
7. Rendle S. Factorization Machines. international conference on data mining, 2010: 995-1000.
8. Song W, Shi C, Xiao Z, et al. AutoInt: Automatic Feature Interaction Learning via Self-Attentive Neural Networks. conference on information and knowledge management, 2019: 1161-1170.
9. Vaswani A, Shazeer N, Parmar N, et al. Attention is All you Need. neural information processing systems, 2017: 5998-6008.
10. Xiao J, Ye H, He X, et al. Attentional Factorization Machines: Learning the Weight of Feature Interactions via Attention Networks. international joint conference on artificial intelligence, 2017: 3119-3125.
11. Zhao G, Lin J, Zhang Z, et al. Explicit Sparse Transformer: Concentrated Attention Through Explicit Selection. arXiv: Computation and Language, 2019.